

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 June 2026
Roll Number	23A51A0535
Project Name	Library Book Request
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

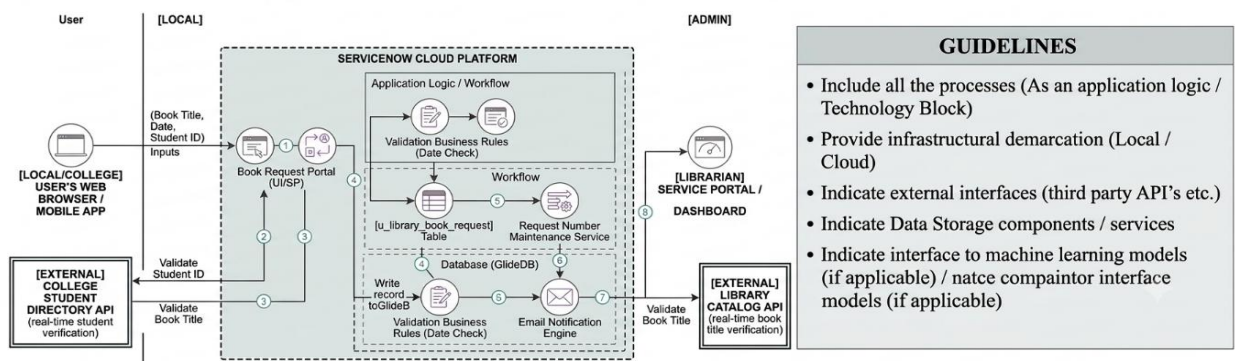


Table 1:

ServiceNow Library Book Request Project: Component and Technology Stack				
S.No	Component	Description	Description	Technology
1.	User Interface	ServiceNow Portal UI	User interacts with ServiceNow Portal	AngularJS / ReactJS / CSS / React. Js, / etc, etc.
2.	Application Logic-1	Custom Table Logic & BRs	Business Rules for validation: Date < current, setAbortAction, clear date, and rtate, etc.	JavaScript / ServiceNow Server-side API
3.	Application Logic-2	Auto-generate Req #	Number Maintenance logic	ServiceNow Number Maintenance
4.	Application Logic-3	Workflow Automation	Approvals & fulfillment process	ServiceNow Flow Designer / Workflow Editor
5.	Database	Request Data & Master Data	The custom request table, book data, user data	ServiceNow Database (MariaDB-based)
6.	Cloud Database	External Catalog Integration	Optional: Syncing with exterior data	IntegrationHub / External APIs
7.	File Storage	Request Attachments	(e.g., scanned IDs or book cover images)	ServiceNow Attachment table [sys_attachment]
8.	External API-1	Student Verification	Integration with college SIS	IntegrationHub / REST / SOAP
9.	External API-2	Book Catalog Sync	(e.g., with a broader ISBN database)	IntegrationHub / external Book APIs
10.	Machine Learning Model	Inventory Prediction	(e.g., predicting book popularity)	ServiceNow Predictive Intelligence
11.	Infrastructure	Application Hosting & Deployment	(e.g., ServiceNow instance management)	ServiceNow SaaS Platform / PDI / Managed Instance

Table 2:

servicenow Library Book Request Based on the non-functional requirements			
S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Leveraging modern front-end frameworks (React, Angular, Vue.js) for a responsive UI.	 React, Angular, Vue.js,  Vue-based Components 
2	Security Implementations	AES-256 Encryption, IAM Controls (Student/Librarian), Firewall Protected, OWASP Best Practices.	AES-256, IAM (RBAC), WAF, OWASP Guidelines
3	Scalable Architecture	Built for growth, handling increasing request volumes smoothly using a 3-tier Micro-services pattern.	3-tier, Micro-services (ServiceNow App Engine)
4	Availability	Ensuring 24/7 access to book request services with load balancer & distributed servers.	Load Balancing, Distributed Servers (ServiceNow platform resilience)
S.No	Characteristics	Description	Technology
5	Performance	Optimized Performance: Fast page load, CDN optimized, cache utilized for rapid form processing.	CDN, Caching (ServiceNow platform and app level)